Age imbalance accounts for the discriminative value of clinical metadata for Parkinson's disease in an open cohort: a leakage-safe evaluation

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Code, data manifests, results tables and figures: https://github.com/EkeminiThompson/neurodegenerative-disease-prediction. All results use the public participant-level tables of OpenNeuro datasets ds007526 (development) and ds002778 (independent test).

Abstract

Background. Machine-learning studies of neurological disease often report performance from cohorts in which diagnostic groups differ in age, and from evaluation designs that can leak participant information across splits. Baselines that show how much of an apparent signal is demographic are rarely reported.

Methods. We analysed the open (CC0) cohort ds007526 (144 participants: 116 Parkinson's disease [PD], 28 healthy controls [HC]). With repeated, stratified, participant-grouped cross-validation and participant-level bootstrap confidence intervals, we trained five classifiers on age, sex and MoCA to separate PD from HC, compared them with paired bootstrap tests, assessed calibration, and repeated the analysis on a 1:1 age-matched subset (n = 54). We also modelled UPDRS part III within PD (n = 113).

Results. In the full cohort, age alone reached ROC-AUC 0.61-0.68 and age + sex + MoCA 0.62-0.71 (best: logistic regression 0.707, 95% CI 0.598-0.802). No model differed from another after Holm correction. In the age-matched subset, every clinical feature set was indistinguishable from chance (all 95% CIs include 0.5 or lie below it; e.g. sex + MoCA, logistic regression, 0.451, 0.291-0.616). A formal paired bootstrap test confirmed that full-cohort discrimination exceeds what the same models achieve among age-matched participants (e.g. sex + MoCA, logistic regression: 0.714 in the full cohort vs 0.567 for the same predictions restricted to matched participants; difference 0.148, 95% CI 0.056-0.252). Calibration was poor (expected calibration error 0.013-0.253). Models fitted on all of ds007526 and applied unchanged to the independent, age-balanced cohort ds002778 (n = 31) gave AUC 0.52 (95% CI 0.30-0.74; permutation p = 0.89) for age alone. Within PD, clinical variables reduced UPDRS-III error from 10.2 to 8.3 points.

Conclusions. In this cohort the discriminative value of basic clinical metadata is explained by the age imbalance between groups. Any model for this cohort should be benchmarked against an age-matched baseline rather than against the full-cohort figure.

1. Introduction

Prediction models for neurological disease are increasingly proposed as markers of diagnosis, but their credibility depends on evaluation design. Information leaking between training and test data inflates performance across many machine-learning-based sciences [2]; cross-validation choices change conclusions in neuroimaging decoding [1]; and best-practice guidance for prediction in psychiatry and neuroscience asks for external validation and explicit confound control [3]. A less discussed but equally basic issue is that diagnostic groups in small open cohorts frequently differ in age, so that a classifier can succeed without using any disease-related information.

This paper provides (i) a **leakage-safe, confound-aware baseline** for the open Parkinson's cohort ds007526, quantifying how much of the discriminative value of clinical metadata is demographic, and (ii) a documented, unit-tested, open implementation: participant-grouped repeated cross-validation, participant-level bootstrap, paired model comparison, calibration, age-matched sensitivity analysis and nested tuning. We do not claim methodological novelty for the individual components.

2. Data

ds007526 (OpenNeuro v1.0.2, CC0, DOI 10.18112/openneuro.ds007526.v1.0.2; Neurology Institute, Tel Aviv Sourasky Medical Center) [4]. Verified from the dataset's `participants.tsv` (SHA-256 and mirror commit in `data/manifests/ds007526.json`): 144 participants (116 PD, 28 HC). Participant-table variables and non-missing counts: age (144), sex (144), MoCA (140), UPDRS part III (139), UPDRS total (139), disease duration (116, PD only; months), levodopa equivalent daily dose (LEDD; 115, PD only). Inclusion criteria for PD were Hoehn & Yahr stage at most 3 and MoCA at least 21, which limits severity range. Healthy controls were younger than patients (mean 60.8, SD 7.5 vs 66.1, SD 9.4 years; difference 5.3 years, Welch p = 0.002); the proportion of women was 54% in HC and 38% in PD (Fisher p = 0.14).

3. Methods

Classification features. Age, sex (F = 0, M = 1) and MoCA. UPDRS, LEDD, disease duration, PIGD and tremor scores were excluded because they are defined only for, or diagnostic of, PD (label leakage). Complete cases: n = 140 (113 PD, 27 HC) for age + sex + MoCA; n = 144 for age alone.

Models. Logistic regression, random forest, RBF support vector machine, XGBoost and LightGBM [6], each in a pipeline of median imputation, standardisation and estimator fitted inside each training fold, with class weighting for the 4:1 imbalance.

Evaluation. Stratified group 5-fold cross-validation with participant as group, repeated 20 times (10 for the extended analyses), with an assertion that no participant appears in both training and test folds. A participant's score is the mean of its out-of-fold predictions across repeats. Confidence intervals are 95% percentile intervals from 2,000 bootstrap resamples of whole participants. Metrics: ROC-AUC (primary), PR-AUC, balanced accuracy, sensitivity, specificity, Brier score and expected calibration error (ECE; 10 bins).

Extended analyses. *A1:* paired participant-level bootstrap of AUC differences for all 10 model pairs, with Holm correction [5]. *A2:* reliability curves, Brier score, ECE. *A3:* every HC was matched 1:1 without replacement to the PD participant nearest in age (caliper 3 years), giving 27 + 27 participants (mean age 61.6 vs 61.3 years); models were re-evaluated with the same protocol. *A4:* out-of-fold performance by sex. *A5:* nested cross-validation (inner grouped 3-fold grid search of logistic regression C and random-forest leaf size and depth inside each outer fold; 3 repeats). **Severity (E1):** ridge and random-forest regression of UPDRS part III on age, sex, MoCA, disease duration and LEDD within PD (n = 113), against a training-mean baseline; MAE and Spearman correlation.

Full-versus-matched comparison (X2). To compare like with like, every feature set of the age-matched analysis was also evaluated in the full complete-case cohort (n = 140; 20 repeats). Because the matched subset is nested in the full cohort, participants were resampled once (5,000 bootstrap resamples) and two paired contrasts were computed on each resample: D1, the AUC of the full-cohort models minus the AUC of models trained and cross-validated inside the matched subset; and D2, the AUC of the full-cohort out-of-fold predictions for all participants minus their AUC restricted to the matched participants (same model, same predictions, different sample composition). Two-sided bootstrap p = 2 x the smaller tail. D2 is the primary contrast, because D1 is exaggerated by the below-chance cross-validation artefact described under A3. X2 is exploratory and was specified after A3.

Independent-cohort test (X1). Logistic regression on age, sex, or both was fitted on all 144 ds007526 participants and applied without refitting to ds002778 (UC San Diego; 15 PD, 16 HC; no shared participants), in which age is balanced between groups (63.3 vs 63.5 years, Welch p = 0.94). MoCA and MMSE are different instruments and were not used. We report AUC, a participant-level bootstrap interval (5,000 resamples) and a two-sided permutation p-value (5,000 permutations). This was specified after A3 and is exploratory.

Order of analyses. E0 (classification) and E1 (severity) were specified and run first. A1-A5 were added after seeing E0 and are therefore exploratory. Seed 20261108; every run appends its configuration and metrics to `reports/results/runs.jsonl`.

4. Results

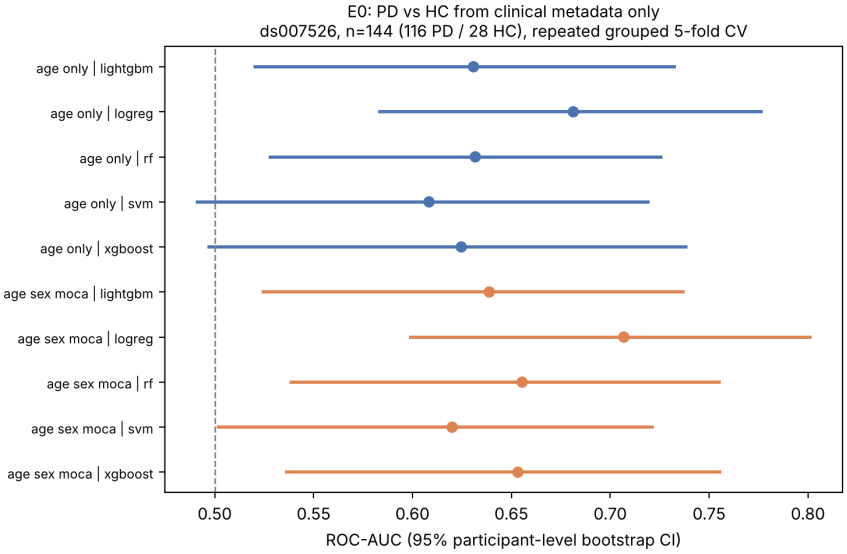


Figure 1: Full cohort: clinical-only ROC-AUC with 95% CI (ds007526, n = 144).

Full cohort (E0). Age alone: logistic regression 0.681 (0.583-0.777), random forest 0.632 (0.527-0.726), SVM 0.608 (0.490-0.720), XGBoost 0.625 (0.496-0.739), LightGBM 0.631 (0.520-0.733). Age + sex + MoCA: 0.707 (0.598-0.802), 0.655 (0.538-0.756), 0.620 (0.501-0.722), 0.653 (0.536-0.756), 0.639 (0.524-0.738), respectively. Adding sex and MoCA changed AUC by at most 0.03 and the intervals overlap throughout.

Model comparison (A1). No pair of models differed after Holm correction (smallest raw bootstrap $p = 0.095$, logistic regression vs SVM; all Holm-adjusted $p \geq 0.95$). Model choice is not a meaningful source of variation on three clinical features.

Age-matched subset (A3). With age balanced, no clinical feature set separated PD from HC above chance:

Features (logistic regression)	ROC-AUC	95% CI
Age only	0.346	0.212-0.505
Sex + MoCA	0.451	0.291-0.616
Age + sex + MoCA	0.433	0.276-0.598

Random forest gave 0.152 (0.051-0.288) for age alone, 0.477 (0.321-0.628) for sex + MoCA and 0.372 (0.232-0.519) for all three. Point estimates below 0.5 are a known artefact of cross-validating noise in very small balanced samples (training folds carry chance imbalances that reverse in the held-out fold); they are not evidence of an inverse relationship and are not interpreted. With 54 participants the intervals are wide (for example 0.29-0.62 for sex + MoCA), so a modest true effect cannot be excluded. The result is that *the full-cohort AUC of about 0.7 is not evidence of disease-related information.*

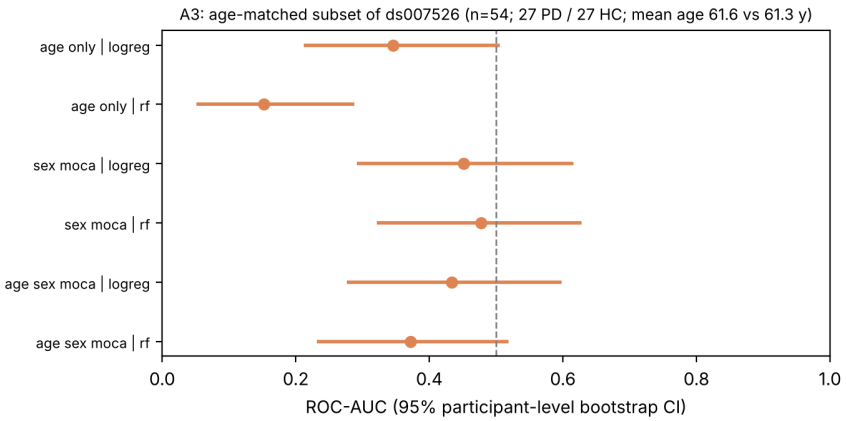


Figure 2: Age-matched subset (n = 54): all clinical intervals include chance or lie below it.

Full versus matched (X2). Like-for-like, the full-cohort AUC of logistic regression was 0.667 for age, 0.714 for sex + MoCA and 0.705 for age + sex + MoCA (n = 140). Restricting the same out-of-fold predictions to the 54 matched participants gave 0.490, 0.567 and 0.556. The paired differences (D2) were 0.177 (95% CI 0.067-0.297), 0.148 (0.056-0.252) and 0.149 (0.055-0.259), all $p \leq 0.001$; for random forest D2 was 0.367, 0.138 and 0.256, with all intervals excluding zero. Against models trained inside the matched subset (D1), logistic regression differences were 0.368, 0.286 and 0.289 (all intervals excluding zero) and the random forest differences were 0.516, 0.115 (CI -0.048 to 0.271) and 0.306. The age-only model, which is the cleanest case, discriminated at 0.667 in the full cohort and at chance (0.490) among age-matched participants. Sex + MoCA also lost most of its apparent discrimination in matched participants, consistent with MoCA and sex acting partly through their association with age or the sampling of controls. The full-cohort estimates here use 20 repeats and the matched-subset estimates in A3 use 10, so matched-subset AUCs differ slightly between analyses (for example sex + MoCA 0.428 here vs 0.451 in A3); conclusions are unchanged.

Independent cohort (X1). In ds002778, age-only AUC within the cohort was 0.517, matching the absence of an age difference. The ds007526-trained models gave AUC 0.517 (0.303-0.737; permutation p = 0.89) for age, 0.513 (0.295-0.723; p = 0.91) for age + sex and 0.515 (0.338-0.690; p = 1.00) for sex, against apparent development AUCs of 0.694, 0.699 and 0.578. The signal that looked like 0.7 in the development cohort did not transfer to a cohort without the age imbalance. With 31 participants the intervals are wide and a modest effect cannot be excluded; the result agrees with, and is independent of, the age-matched subset in ds007526.

Calibration (A2). ECE ranged from 0.013 (SVM) to 0.253 (logistic regression); Brier scores 0.151-0.217. The SVM's low ECE reflects predictions close to the 81% PD prior for nearly everyone (specificity 0 at threshold 0.5 in E0), not discrimination. None of these models yields interpretable risks.

Sex subgroups (A4, exploratory). Apparent discrimination was concentrated in women (logistic regression 0.756, 0.612-0.875; 15 HC) and absent in men (0.600, 0.439-0.772; 12 HC); tree models were at or below chance in men (e.g. random forest 0.481, 0.301-0.660). With 12-15 controls per sex this is hypothesis-generating only.

Nested tuning (A5). Logistic regression 0.681 (0.565-0.783) and random forest 0.705 (0.596-0.803), n = 140; tuning did not change the conclusions.

Severity within PD (E1). Training-mean baseline MAE 10.16 points (8.81-11.57); ridge 8.26 (6.93-9.69; Spearman 0.463); random forest 8.60 (7.29-9.94; Spearman 0.456). The improvement is about 2 UPDRS points with overlapping intervals.

5. Discussion

What the evidence supports. In this open cohort, basic clinical metadata separates PD from HC only to the extent that age differs between groups: the full-cohort AUC of 0.61-0.71 collapses to chance once age is balanced. The finding is modest, specific and reproducible from the released code, and it held in an independent,

age-balanced cohort. Its practical consequence is a benchmark: models for this cohort should be reported with an age-matched analysis, and full-cohort AUCs near 0.7 from three demographic variables show how easily such a figure can be reached without any disease-related information.

Evaluation lessons. Three results transfer beyond this cohort: model choice was irrelevant on low-dimensional inputs (no pair differed), nested tuning did not alter the conclusions, and threshold-based metrics and calibration were unstable at an 81% prevalence, so AUC with participant-level intervals was the only stable summary.

Limitations. Development in one cohort from one site, with a single small independent test cohort (n = 31) that used a different site and protocol; small control group (27-28 HC) and imbalanced classes; A1-A5 are exploratory; the age-matched analysis has low power and wide intervals; unknown medication state at assessment; narrow severity range (Hoehn & Yahr at most 3, MoCA at least 21); sex-subgroup estimates rest on 12-15 controls. The models are research baselines, not clinically validated tools.

6. Data and code availability

All code, manifests (dataset version, mirror commit, checksum), configuration, result tables, figures and a JSONL run log are in the repository. Reproduction: `python scripts/fetch_metadata.py`, `python scripts/run_clinical_baseline.py --repeats 20`, `python scripts/run_clinical_extended.py --repeats 10`, `python scripts/run_matched_vs_full.py`, `python scripts/run_clinical_external.py`, `python scripts/make_figures.py`. Continuous integration runs the unit tests and linting on hand-built fixtures only.

Declarations

Ethics. This is a secondary analysis of de-identified, openly licensed (CC0) participant-level tables from OpenNeuro datasets ds007526 and ds002778. No new data were collected and no additional ethical approval was sought.

Competing interests. The author declares no competing interests.

Funding. This work received no specific funding.

Data availability. All data are public on OpenNeuro (ds007526 v1.0.2, doi:10.18112/openneuro.ds007526.v1.0.2; ds002778 v1.0.5). Code, manifests, results tables and figures are at https://github.com/EkeminiThompson/neurodegenerative-disease-prediction (archived at doi:10.5281/zenodo.24102101).

AI usage. An AI assistant (Anthropic's Claude) was used to help write the analysis code, run the analyses and draft the text. The author directed the work and is responsible for the content, and all reported numbers come from the released code and data.

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